

What you get on 27 October

A **works-level dataset over a frozen frame of 4,299,418 works Canadian by any of four checkable metadata routes**, in which every record carries: **why it was retrieved** (route provenance), **which sense of “Canadian” it satisfies**, language, its **categories** (multi-label, each cited to its own literature), the **Ioannidis domain**, **data and code links**, and, per field, a **validation status**: human-verified, gated label, or **score only**. On it, the summaries a landscape is for: **counts and trends by category, year, language, province, institution and funder, with design-based intervals**. With it: the **cited inclusion criteria and the census they came from**; a **preregistered bilingual human audit** giving prevalence, sensitivity and **differential error by language, source and abstract availability**; every prompt, label and disagreement; and a **bilingual explorer, live at `metacan.xera.ac`**.

The frame is **built, not sampled**: all **482 partitions** of a pinned OpenAlex snapshot, each work once. The pilot below is done and re-derivable offline.

Don’t search for metaresearch. Search for Canada, then screen.

Retrieve what *looks* like metaresearch and the field’s boundary becomes a property of your keyword list, and the map is unauditable. I inverted it: the frame is **Canadian research as four checkable metadata routes see it** (affiliation, funder country, venue, aboutness), owing nothing to my notion of metaresearch, so membership becomes a **classification** over that frame, not a **retrieval** over the literature. Retrieval fails both ways, measured. The best topic route recovers **12% of the works our screen calls metaresearch** (7% under the other screener), because OpenAlex files a work by what it is *about*: metaresearch about cardiology reads as cardiology, so **the problem is aboutness, not vocabulary**. And it wrongly *admits*: **17,466 works** enter on six **suspect institution assignments** (parse failures, real organizations on unrelated papers), flagged for audit. **1,565,226 works (36.4%) carry no Canadian affiliation at all**, so an affiliation-only frame never sees them.

The criteria, derived rather than asserted

Three frontier models screened the same **5,600 works**, drawn with known probabilities, under one locked rubric.

Rate agreement is not set agreement. Of the **274 works** any model called metaresearch, only **104 (38%)** were called metaresearch by **all three**; **117 (43%)** rest on one model’s opinion; pairwise overlap is about **50%**. At a ~1% base rate the settled rejects buy near-total agreement for free. **The boundary is not a line the models share; it is a region they each cut differently**, stable at $n = 1,000, 2,000$ and $5,600$.

The field has no consensus definition either, and says so in print: Puljak et al. analysed **175 information sources**, found “*definitions varied*”, and called for consensus (J Comp Eff Res 2020); Kataoka et al. **publicly dispute them** (2023;154:219); there is **no MeSH heading** for meta-research (Stevens & Laynor 2023). Those are claims about humans, mine about models under one rubric; **only the human audit closes the gap**.

So I made the *why* measurable. A separate masked adjudicator (**a fourth model, not one of the arms**; opinions shown as A/B/C, no model names) adjudicated **all 179 contested works**, naming **which seam produced each split**. No arm preference is detected ($p = 0.49$). Its verdict: **the rubric was silent on 159 of them (89%)**; only **17 splits** were a screener misapplying a rule.

That turns adjudication into a **frequency-weighted census of the sentences the rubric was missing**. I published the **census, wrote the revision against it, and locked it before re-screening**, so it could not be tuned to the numbers. **Re-screened: 96 of 179 (54%) are now unanimous; pairwise overlap moves 0.19 to 0.36**. Those works were *selected for disagreement*, so the full re-screen is the unbiased test. **These are the call’s “inclusion and exclusion criteria”, written against measured evidence of where they fail rather than asserted, and the community’s own criteria are what the audit and the released disagreements put up for challenge**. The residual **83 works** ship contested.

1. Methodology

No category is defined from memory. Six are **quoted verbatim** from a source that defines the field, committed to the repository: *metaresearch* (Ioannidis, PLoS Biol 2015;13:e1002264, his five areas becoming a per-record domain); *bibliometrics* (Mingers & Leydesdorff 2015); *sts* (Jasanoff 2004; Latour); *scholarly communication* (Borgman 2007); *open science* (UNESCO 2021); and *metaepidemiology*, coded under **both** published definitions (Murad 2017; Kataoka 2023) and **flagged where they disagree**, because a live dispute is a boundary to be *measured*, not settled. **The seventh is weaker and the rubric says so:** Fanelli and COPE define *misconduct* and *publication ethics*, **not the field that studies them**, so *research_integrity* is defined by its **object** and the gap is on the record. **The screen is multi-label:** a bibliometric study of citation distortion is *both*, and forcing the choice was the bug. **“Adjacent” was bounded by negation**, its edge *not-core*, which is why **OUT-vs-adjacent was our largest confusion at every sample size**. The build **fails** if the schema allows a category the rubric never defines, or if a quotation marked verbatim is not a substring of its cited PDF (it caught one: v3.0 quoted Murad with a word he did not write, D29).

Estimand, prespecified. *Primary:* Canadian-produced metaresearch (≥ 1 Canadian affiliation or funder); **71.2% of the frame carries no funder metadata**, so **both clauses of the primary** rest on sparse metadata, and every record carries **provenance**. *Secondary:* metaresearch **about** the Canadian research system, **prespecified but pilot-constrained**: the route named “about

Canada” and the rubric’s *about the research system* are **different constructs** (only **39 of 5,600** works are both in scope and about Canada), so v3.1 **splits the field in two** and the secondary estimate ships only if the redesigned fields survive the audit. **44,008 paratext and short-title records (1.0%) are excluded by declaration**, shipped flagged, outside every estimate.

What broke, and what now runs. Design weights survive a noisy stratifier **only while every record keeps a nonzero selection probability**, and **my own design violated exactly that**: I never checked $\text{sum}(N_h) = N$, so **549,370 works (12.9%) had inclusion probability zero**, including **328,912 in the secondary estimand’s own cell**. Adversarial review found it; seven strata now **partition** the frame, asserted on every build. *DEVIATIONS.md* **carries all thirty-three entries; every one produced output that looked correct, and none threw**.

One annotation contract, and no layer borrows another’s authority. The LLM reads **every** work under the full v3.1 rubric (**\$1,261**: the screen is affordable, so there is no prefilter and no distilled substitute), and that output is **provisional, machine, per-field**. The classifier trains on those labels and **never asserts a category**: distilled from our *metaresearch* labels it reproduced **its own teacher’s positive set at Jaccard 0.17**, a fact about the *boundary* rather than an error rate, and disqualifying either way. What it is *for* is what another LLM pass cannot buy: a **calibrated score** the audit stratifies on, a **frame-wide map of where the three models would disagree** (running all three over 4.3M costs 3× and 25 days; a 5,600-work sample predicts it), and **rubric revisions testable in minutes, not one \$1,261 pass each**. **Labels ship only where checked**: human-verified, or gated against MEDLINE publication types by NLM’s **record-level** *IndexingMethod* rather than a year cutoff, with **class-specific calendar eligibility**, because *Observational Study* enters the vocabulary in 2014 and *Systematic Review* in 2019, so their absence before that is **not a negative**. Everything else is a **score**. **The dataset states, per field, which annotations it has earned the right to assert**.

Validation, because a simple sample of the 4,243,096 rejects returns an expected **0.4** misses. **(i) A stratified probability sample with floors** (machine score, **between-model disagreement**, language, source, abstract availability), drawn from **every** stratum at known nonzero probabilities; two humans code independently, blind to machine labels, **on the full-text cascade rather than the models’ own payload** (blinding does not create missing information); an **unresolved outcome exists** and estimates are **conditional on ascertainment**. **(ii) Known-item recall on external criteria**: venue, registries, MEDLINE types, all immune to aboutness.

Sample: $n \approx 1,000$, dual-coded. **The second coder is the largest delivery risk and is not optional**: recruited in Week 1 via CaRN, ACFAS, CAIS-ACSI and CARL; **if none is found the French claim is withdrawn, not fudged**. **Costs, after costing them wrong three times (D7, D30): \$1,279 all in**. **The call’s CAD \$4,000 covers travel and participation, in its own words, so no research cost assumes it**: compute is self-funded, and the coder is paid from the award only if the organizers confirm eligibility, otherwise via in-kind support.

2. Openness, transparency, reproducibility

The repository is **public** and the committee can run it today. *make pilot-offline* re-derives **every figure above** from the archived artifacts **with no network** (frame inputs come from a **pinned S3 snapshot**, not the metered API’s 8.2-day pass); *make proposal* **fails if this page quotes a number the pilot does not produce or declare**; *make lint* fails if the strata do not partition the frame, the codebook contradicts its schema, or a quotation drifts from its source. **Each guard exists because that failure happened**, the newest of them after a reviewer noticed this page claimed a check nobody ran (D31).

One reproducibility claim I will not make: temperature 0 does not make an LLM deterministic and these versions will be retired, so I release the **exact prompts and raw outputs** but do **not** claim the labels reproduce. Each release carries a Zenodo DOI: data (CC-BY-4.0), code (MIT), Docker, lockfiles, **every rubric version, labels and disagreements**, the **seam census**, validation results *with the errors*, a **Datasheet**, *DEVIATIONS.md*. Preregistered on **OSF**.

3. Diversity, assumptions, limitations, bias

I am a solo applicant working in English. I do **not** claim to represent the field’s diverse traditions; that needs collaborators I do not have, and I once claimed to have *measured* how badly I serve them and **withdrew it** (four records, $p = 0.141$). What I could stop doing was defining anyone by their distance from me; §1 is that repair.

Érudit, the main francophone Canadian platform, matches 0 OpenAlex sources by name (OAI-PMH live, 379 sets); whether OpenAlex holds its corpus under other source records is **unmeasured until the ISSN/DOI crosswalk runs (W3)**, and the verified harvest supplies the rest. Funder metadata skews against the traditions the call asks me to include (**NSERC carries 9.2× SSHRC’s**), and **23.3% of the frame (1,003,117 works) has no abstract**, where the pilot screen finds half as much metaresearch ($p = 0.023$).

Assumptions named. The base rate rests on machine labels: **a hypothesis with a denominator, not a result**, and coverage holds *within the frame*. No sensitive identity is inferred; no derived labels where Indigenous-governed data are implicated; OCAP® without an Indigenous partner is compliance theatre. I submit as an independent researcher with no affiliation: **one of the people this dataset cannot see**.

4. Work plan (1 Aug to 27 Oct)

W1–2 Preregister; **recruit the coder**. · **W3–5** Érudit harvest; full-frame screen; version difference. · **W6–8** Stratified sample, in and out; dual blinded coding; adjudication. · **W9–10** Estimates; bias report; MEDLINE validation. · **W11–12** Release; deploy; plenary talk.